

Kaggle - Random Forest #08



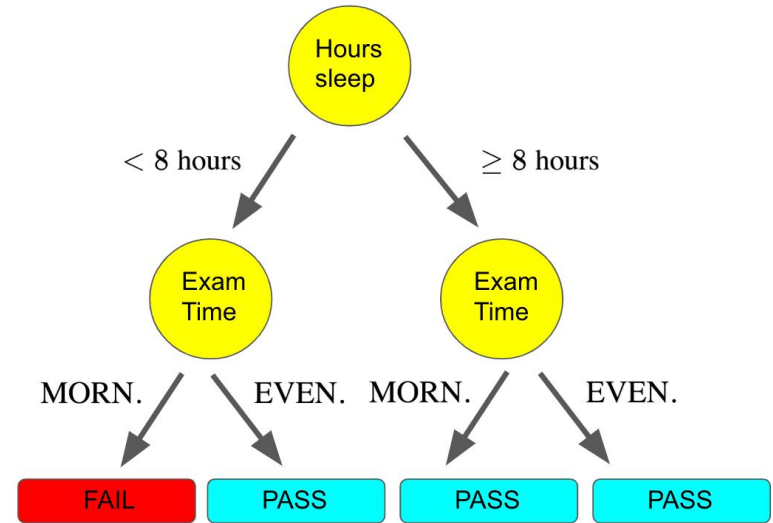
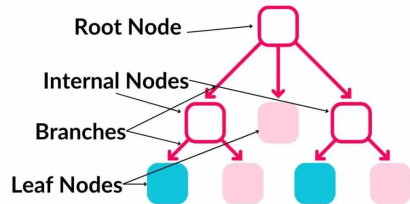
Recap of Kaggle so far

- 08/10 – Kaggle 01: Welcome to QMML! + EDA w/ Maiesha
- 15/10 – Kaggle 02: Pandas, Matplotlib w/ Maiesha, Monesh & Girish
- 22/10 – Kaggle 03: ML Modelling w/ Stefan
- 29/10 – Kaggle 04: Hyperparameter Tuning w/ Girish
- 05/11 – Kaggle 05: Boosting w/ Monesh & Maiesha
- 12/11 – Kaggle 06: Logistic Regression w/ Monesh & Maiesha
- 19/11 – Kaggle 07: Cross Validation w/ Girish

This Month's Kaggle competition (November 2025, S5E11): <https://www.kaggle.com/competitions/playground-series-s5e11>

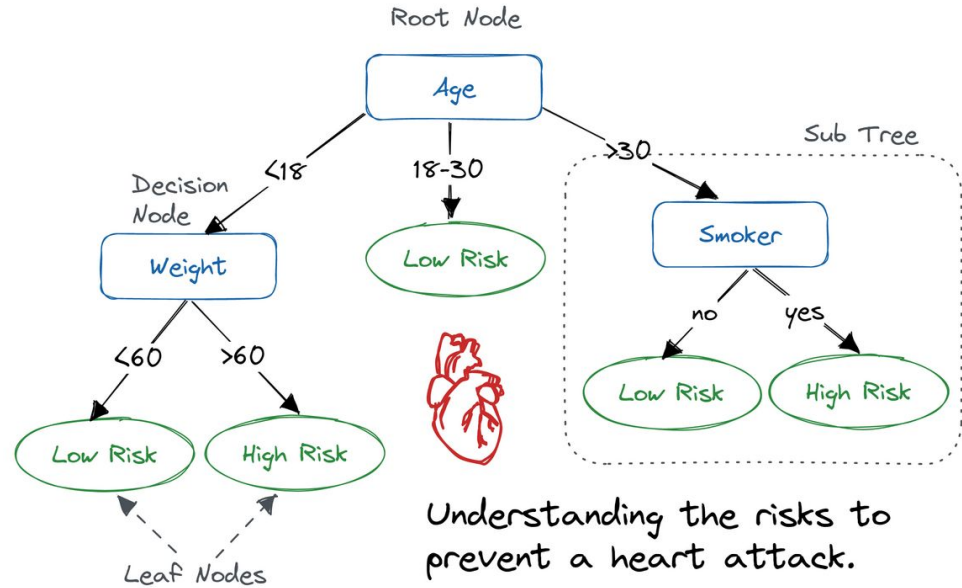
Decision Trees - Recap

- A supervised learning model that uses a series of if/else rules to make predictions
- Splits data into branches based on the feature that best separates the target



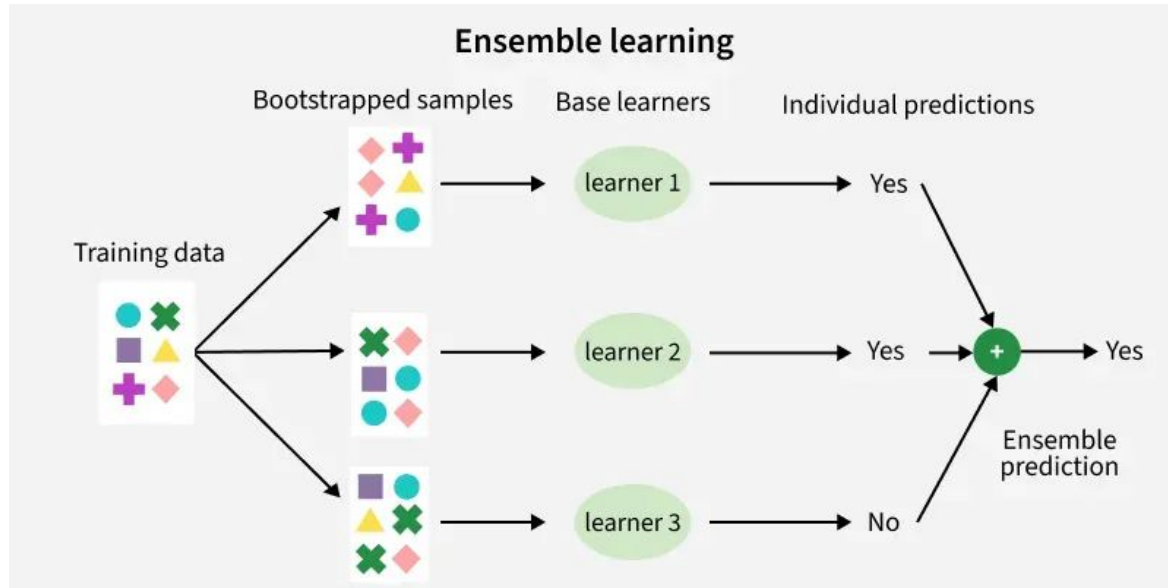
Decision Trees - Recap

Age	Weight	Smoker	Risk level for heart disease
17	45	No	Low risk
30	85	No	Low risk
35	100	Yes	High risk
80	60	No	Low risk



Ensemble Methods - Recap

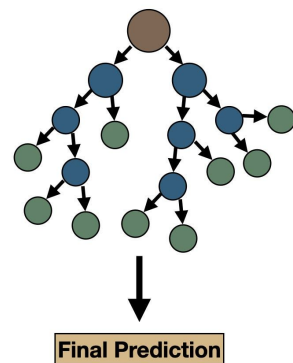
- Machine learning technique that combines multiple individual models to create a single, more accurate, and robust predictive model



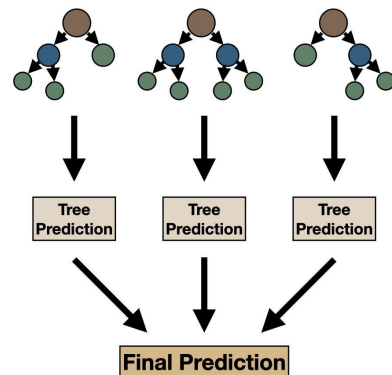
Random Forest

- A machine learning algorithm that uses many decision trees to make better predictions
- For supervised learning
- Ensemble method (when you combine models/ their predictions together)

Single Decision Tree



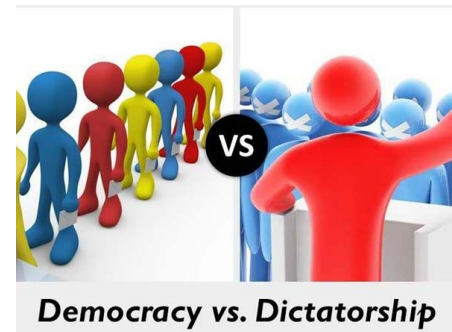
Decision Tree Ensemble



Random Forest

Analogy -

- A decision tree is like a single expert who makes a decision by asking a series of questions
- A random forest is like a group of experts (each with slightly different information) who then vote on the best answer



Step by step how does it work?

1. Select a subset of data points and a subset of features to construct each decision tree
2. Construct multiple decision trees from each sample of the data
3. Each decision tree generates an output : Using majority voting for classification and Averaging for Regression we have a final output

1.From a Dataset create samples

Will I play tennis ?

Day	Outlook	Temperature	Humidity	Wind	Play?
1	Sunny🌞	Hot🔥	High	Weak	No😞
2	Sunny🌞	Hot🔥	High	Strong	No😞
3	Overcast	Cool🧊	Normal	Weak	Yes😊
4	Rain🌧️	Mild😊	Normal	Weak	Yes😊
5.	Rain🌧️	Cool🧊	Normal	Strong	No😞

Bootstrap Sampling : Create multiple datasets by sampling rows with replacement so each sample has the same size as original dataset

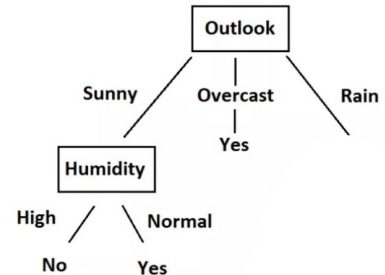
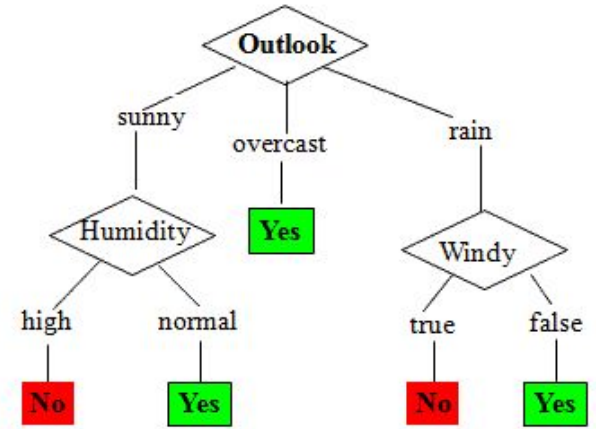
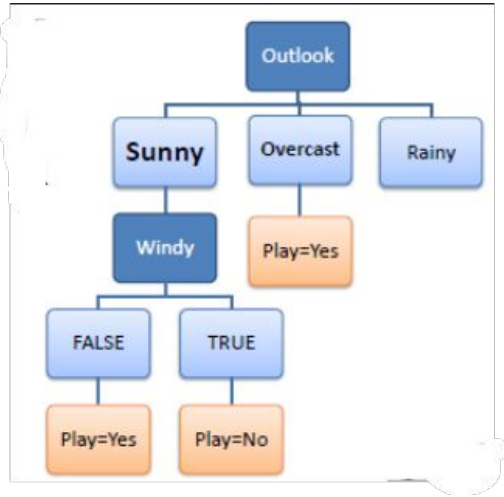
Sample 1: [1,3,2,4,5...]

Sample 2: [4,5,2,1,3..]

Sample 3: [3,4,2,1,5...]

2. Construct Multiple Decision Trees from each sample

For each split in a tree, Random Forest randomly picks a subset of weather features (here 2 out of 4) to consider, making each tree focus on different aspects of the data.

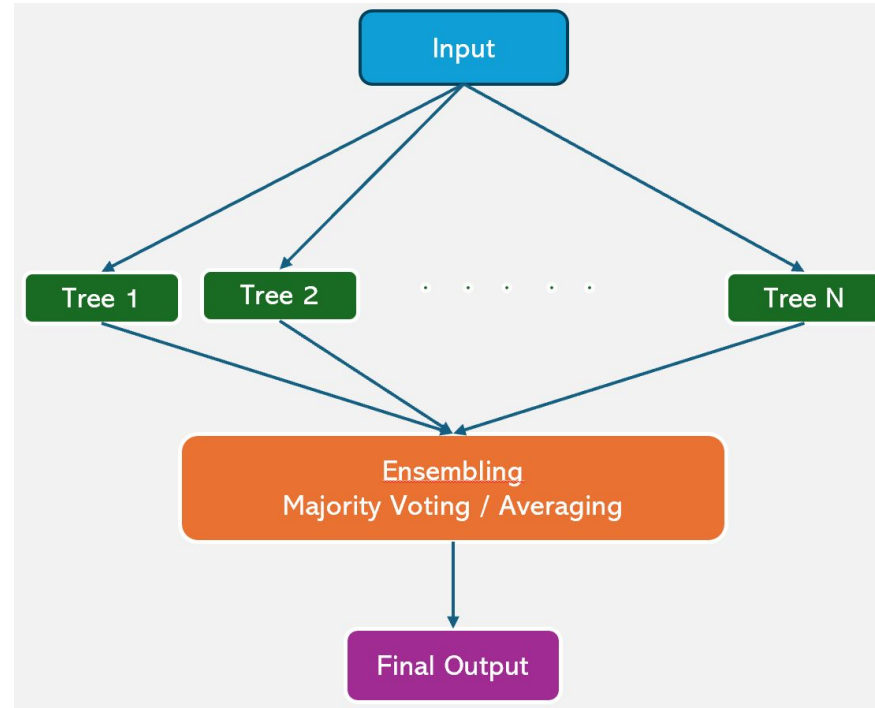


3. Final Predictions/ Classifications

Final Prediction: All trees vote together for the final prediction.

For Classification: Take the Majority vote from all the trees

For Regression: average the predicted values from all the trees



Advantages + Disadvantages

Advantages	Disadvantages
- High accuracy: The ensemble/combination of multiple decision trees typically leads to more accurate predictions than a single tree.	- Computationally expensive: With more trees -> the more memory and computational power is needed
- Prevents overfitting: By averaging the results of many trees, the model is less likely to overfit the training data.	- “Black box” nature it’s difficult to understand how a specific prediction was reached
- Handles missing data: Is robust to datasets with missing values without needing imputation.	- Potential for overfitting
- No data normalization needed: Does not require features to be scaled or normalized.	- Generating predicting can be slow: Each tree must generate its own prediction first before being combined

Important hyperparameter for Random Forest

1. **n_estimators**: Number of trees built before averaging predictions.
2. **max_depth**: maximum depth of the tree
3. **max_features**: Maximum number of features considered for splitting a node.
4. **min_sample_leaf**: Minimum number of leaves required to split an internal node.
5. **max_leaf_nodes**: Maximum leaf nodes allowed in each tree.

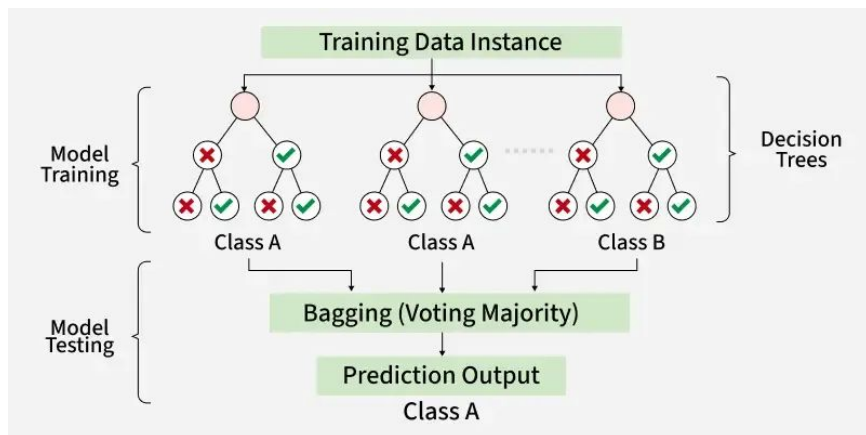
More Resources on Random Forest for Deep Dive

Medium Article:

<https://medium.com/data-science/random-forest-explained-a-visual-guide-with-code-examples-9f736a6e1b3c>

StatsQuest (Youtube): Random Forest playlist

Summary



WHAT IS RANDOM FOREST?

Random Forest is an ensemble learning method that combines multiple decision trees to improve accuracy at prevent overfitting and regression tasks.



HOW DOES IT WORK?

1. **Bootstrapping:** Random subsets of the data are selected (with replacement)
2. **Tree Building:** Multiple decision trees are built independently on these subsets
3. **Random Feature Selection:** Each tree only considers a random selection of features when splitting nodes
4. **Voting/Averaging:** Classification → Majority vote
Regression → Average of outputs



WHY USE RANDOM FOREST?

- Reduces overfitting
- Handles missing values
- Works well with large datasets
- Provides feature importance



APPLICATIONS

- Fraud Detection
- Medical Diagnosis
- Stock Market Prediction
- Customer Segmentation
- Image and Speech Recognition

KEY PARAMETERS

n_estimators: Number of trees
max_depth: Max depth of each tree
max_features: Number of features to consider when looking for the best split
bootstrap

RANDOM FOREST = MULTIPLE WEAK LEARNER → STRONG LEARNER

Feedback Form

